

Srikha Chaganti

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AI/ML researcher building 5X error reducing end-to-end models, with a powerful mix of research, math, core AI frameworks and soft skills

EDUCATION

Master of Science in Applied Mathematics May 2028
Columbia University
Bachelor of Science in Mathematics (GPA: 3.85/4.0) May 2026
University of Texas at Austin - **Dean's Scholar & University Honors**
Honors Thesis: "Unknotting Number under Connected Sum and Knot Mutation" [\[pdf\]](#) advised by Dr. Maggie Miller

PROFESSIONAL EXPERIENCE

ML Engineering Research Intern National Science Foundation Funded REU Jun 2025 – Aug 2025
Illinois Institute of Technology

- Built a novel end-to-end machine learning pipeline for Stress and fatigue classification from physiological sensor data for real-world deployment in construction worker safety monitoring
- Evaluated multiple model architectures with hyperparameter tuning, novel uniLASSO regularization, and prioritized model interpretability to achieve accuracy through reduction in error by 5X
- Co-authored paper (in publication process): "Simulated HRV Analysis for Stress and Fatigue Detection in Construction Workers" [\[pdf\]](#); presented at JMM Conference 2026 and IIT SURE Final Conference

Systems and Signal Processing Research Intern Summer Research Program Jun 2024 – Aug 2024
Fields Institute for Research in Mathematical Sciences

- Constructed a new 'Phase Coherence' metric for acoustic signals enabling new quantitative analysis of otoacoustic emissions
- Building, cross-validation, and optimization of coupled oscillator models against raw, large-scale signal datasets
- Applied nonlinear dynamical systems analysis (Volterra Series) to characterize Blackbox system behavior and invited to present a technical lecture to York University
- Contributed to publication "Calculating Spontaneous Otoacoustic Phase Coherence" (Refer to doi: 10.1121/10.0035326)

Computational Data Analysis Researcher Jun 2023 – Mar 2024
Geometry of Space Laboratory — UT Austin

- Utilized computational geometry and Topological Data Analysis to extract features and map multidimensional, noisy data structures to further understanding of large-scale universe structure
- Built data visualization and processing pipelines to handle, clean, and interpret large-scale spatial distribution datasets
- Wrote survey of literature and paper detailing the process, "Using Genus Statistic Methods to Explore Two Dimensional Topology of Galaxies" [\[pdf\]](#)

TEACHING

Academic Instructor	Berkeley2 SAT Prep Academy	May 2026 - Present
Teaching Assistant	UT Austin Mathematics Department	Aug 2024 – Aug 2025
Content Tutor	UT Austin Texas Athletics	Aug 2023 – Aug 2024
Founder	Lucent Learning	Jun 2020 – April 2025

Founded independent tutoring business; designed personalized curricula for students across introductory through multivariable calculus and managed client communications, marketing collateral, and biweekly progress reporting for 20+ students

TECHNICAL SKILLS AND KNOWLEDGE

Core Domain Expertise: Machine Learning, Supervised / Unsupervised Learning, Signal Processing, Statistical Modeling, Data Pipelines, Hyperparameter Tuning, Predictive Modeling, Prompt Engineering, Data Visualization
Programming Languages: Python, R, SQL, Java, MATLAB, Stata, LaTeX
Libraries and Frameworks: pandas, NumPy, scikit-learn, matplotlib, SciPy, TensorFlow, PyTorch
Expertise in Coursework: Stochastic Processes, Real and Complex Analysis, Topology, Economic Statistics, Genetics, Physics, Language and the Brain
Data Process: ETL, Preprocessing and Modeling, Git, MLflow
Attributes: Critical Thinking, Cross-functional Collaboration, Explainable (Responsible) AI practices
Certifications: BCG Data for Decision Makers Job Simulation on Forage, Anthropic: Claude Code 101

PAPERS & PUBLICATIONS

- *"Unknotting Number Under Connected Sum and Mutation."* [\[pdf\]](#) Presented at UT Austin 2026 Honors Thesis Symposium
- *"Simulated HRV Analysis for Stress and Fatigue Detection in Construction Workers."* [\[pdf\]](#) In publication progress. Presented at Joint Mathematics Meeting 2026 (poster) & IIT SURE 2025 (lecture)
- *"Calculating Spontaneous Otoacoustic Phase Coherence."* Publication. Refer to doi: 10.1121/10.0035326. Presented at Fields Institute FUSRP 2024 Conference
- *"Two-Dimensional Topological Data Analysis on Galaxy Distributions."* Presented at 2024 UT Austin Undergraduate Research Forum; [\[poster\]](#)
- *"Help, Am I hurt?": A grant proposal for External Injury Diagnosis And Identification With Convolutional Neural Networks*
- *"The Illogicality of Artificial Intelligence in the Scientific Research Community"* [\[pdf\]](#)
- *"The Feasibility and Necessity of 100% Renewable Energy Sources"* [\[pdf\]](#)

EXTRACURRICULAR

- Writer, Creative Direction for SPARK, an editorial fashion magazine
- Hair and Makeup Creative Director for Saffron, an editorial fashion magazine
- Multiple award-winning orchestral violinist

Aug 2023 – May 2026

Jan 2023 – Aug 2025